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Assignment: GATE Question 11
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GATE Question 11 Analysis and Hardware Verification

Question 11

Match the logic gates in Column A with their equivalents in Column B.

Column-A	Column-B
1. AND Gate	P. XOR Gate
2. OR Gate	Q. XNOR Gate
3. NOT Gate	R. NAND Gate

Question Analysis

- AND gate gives output 1 only when both inputs are 1.
- OR gate gives output 1 when at least one input is 1.
- NOT gate inverts the input.
- NAND is complement of AND.
- NOR is complement of OR.
- XOR gives output 1 when inputs are different.
- XNOR gives output 1 when inputs are equal.

Thus logical equivalences are verified using Boolean algebra and truth tables.

Boolean Expressions

- AND Gate: $Y = A \cdot B$
- OR Gate: $Y = A + B$
- NOT Gate: $Y = \overline{A}$
- NAND Gate: $Y = \overline{A \cdot B}$
- NOR Gate: $Y = \overline{A + B}$
- XOR Gate: $Y = A \oplus B$
- XNOR Gate: $Y = \overline{A \oplus B}$

Truth Tables

AND Gate

A	B	Output
0	0	0
0	1	0
1	0	0
1	1	1

OR Gate

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	1

NOT Gate

A	Output
0	1
1	0

NAND Gate

A	B	Output
0	0	1
0	1	1
1	0	1
1	1	0

NOR Gate

A	B	Output
0	0	1
0	1	0
1	0	0
1	1	0

XOR Gate

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	0

XNOR Gate

A	B	Output
0	0	1
0	1	0
1	0	0
1	1	1

Arduino Code for Seven Segment Display

```
#include <Arduino.h>

void sevenseg(int a,int b,int c,int d,int e,int f,int g)
{
  digitalWrite(2, a);
  digitalWrite(3, b);
  digitalWrite(4, c);
  digitalWrite(5, d);
  digitalWrite(6, e);
  digitalWrite(7, f);
  digitalWrite(8, g);
}

void setup()
{
  for(int i=2; i<=8; i++)
  {
```

```
    pinMode(i, OUTPUT);
  }
}

void loop()
{
  sevenseg(1,0,0,1,1,1,1);
  delay(1000);

  sevenseg(0,0,1,0,0,1,0);
  delay(1000);

  sevenseg(0,0,0,0,1,1,0);
  delay(1000);
}
```

Code Uploading Steps

1. Create PlatformIO project.
2. Write code in main.cpp.
3. Run pio run to generate hex file.
4. Copy hex file to ArduinoDroid.
5. Connect Arduino using OTG.
6. Upload using Upload Precompiled option.
7. Verify seven segment output.

Hardware Output Verification

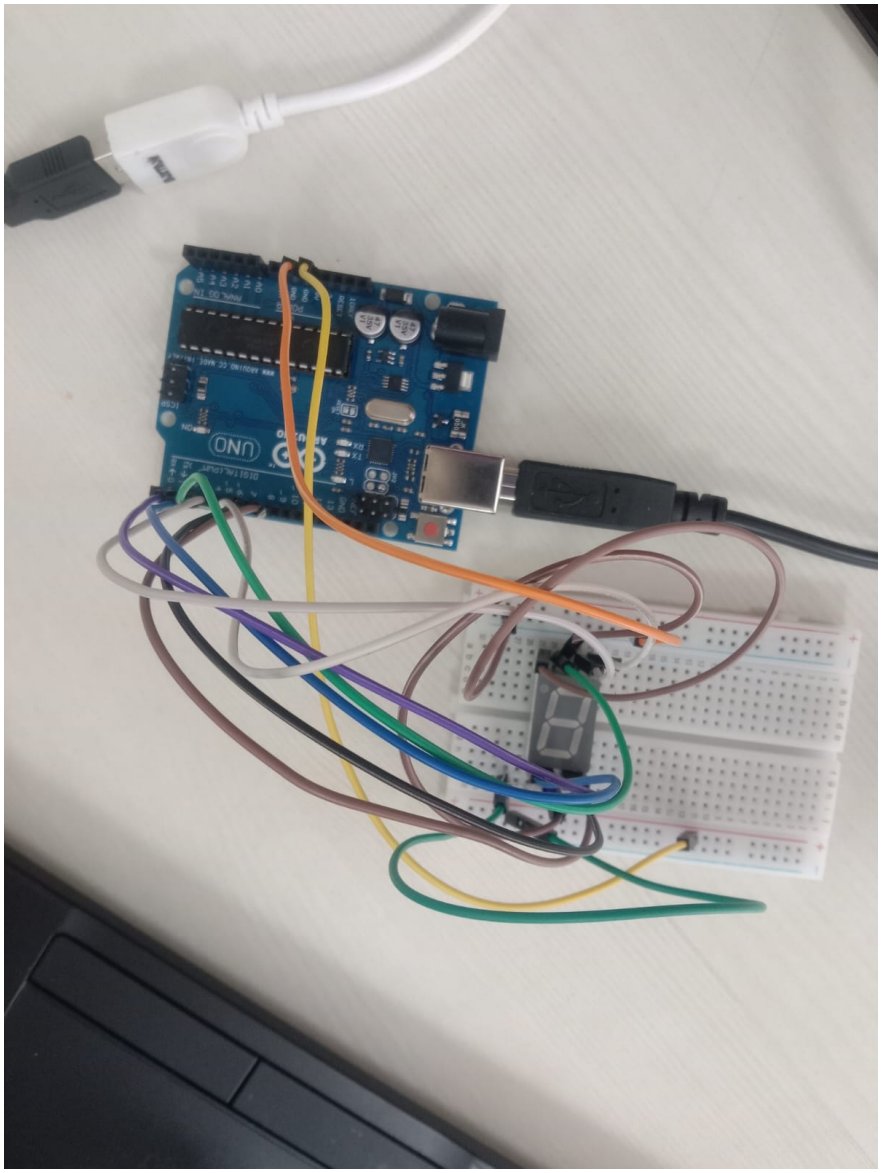

Figure: Seven Segment Display Output

Conclusion

All fundamental logic gates and derived gates were analyzed using Boolean algebra and verified through complete truth tables. The theoretical study was implemented practically using Arduino UNO and a seven segment display. The experimental output confirms correctness of digital logic principles.